

NBA Game Outcome Predictor

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Problem statement

- Predict NBA game outcomes using ML
- My goal: predict + explain why
- Focus on interpretable insights

Data/ Features

Datasets used: Kaggle NBA game dataset

Data:

- Scores
- Shooting %
- Rebounds
- Assists

Features

- FG% diff
- FT% diff
- Rebound diff
- Assist diff

Approach

- Used publicly available datasets
- Cleaned data and add target variable, home_win
- Feature engineering for differences in home and away team stats (FG%, rebounds, assists)
- Logistic regression model
 - Binary classification problem, where the goal is to predict whether the home team wins or loses
- Output
 - Prediction + explanation

Challenges

- I was originally using same game statistics since the data was readily available in the dataset
- Improvements:
 - Switched to using team average statistics, which are available before the game and make the model more realistic

Results

	Feature	Coefficient
0	fg_pct_diff	22.847399
1	ft_pct_diff	3.727890
3	ast_diff	0.092963
2	reb_diff	0.078804

Predicted Winner: Warriors

Win Probability: 0.664

Why this team is favored:

- Warriors have better shooting efficiency.
- Warriors have better free-throw shooting.
- Lakers have a rebounding advantage.
- Warriors have better ball movement based on assists.
- Warriors may benefit from home-court advantage.

Progress

- Data cleaning completed
- Feature engineering done
- Model trained and evaluated
- Improved model implemented
- Explanation system built
- Interactive prediction working
- Working on making it Streamlit